NGSEP 4: EFFICIENT AND ACCURATE IDENTIFICATION OF ORTHOGROUPS AND WHOLE-GENOME ALIGNMENT

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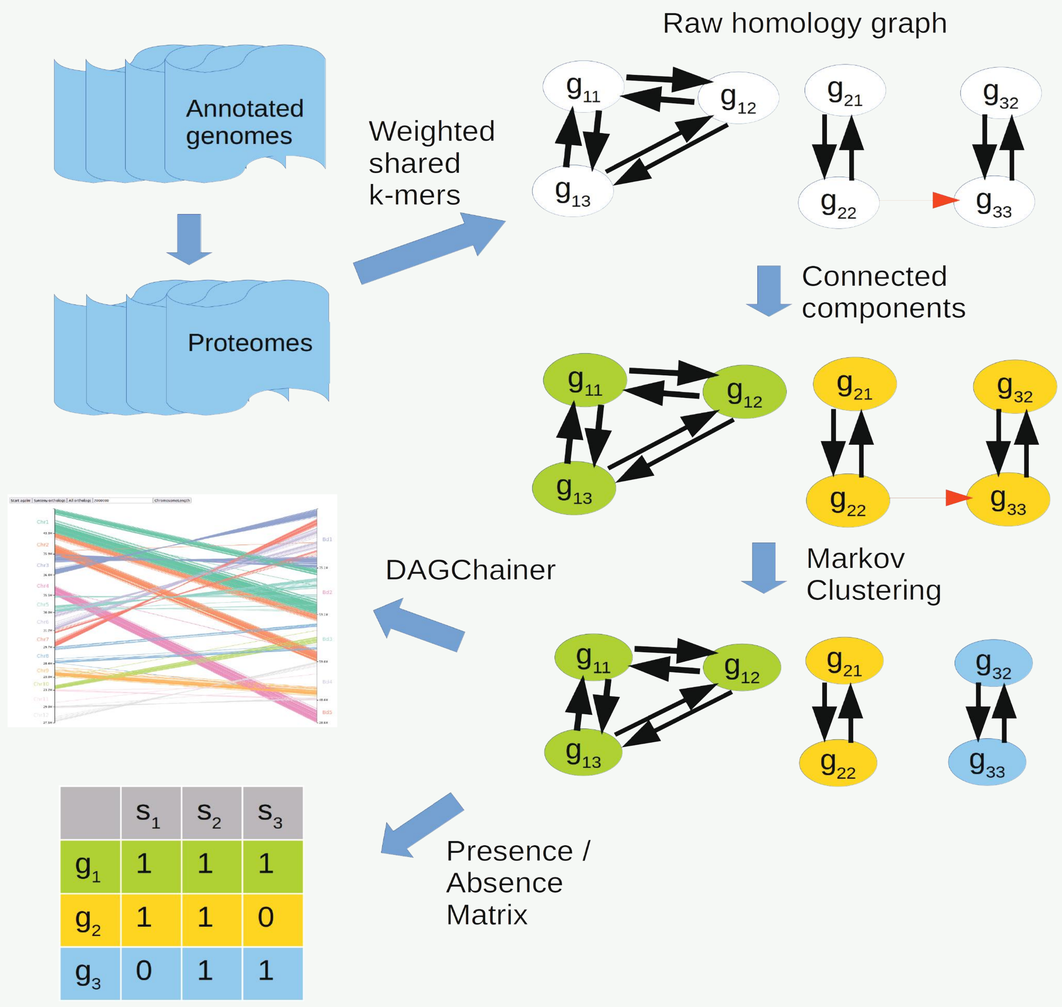
Whole-genome alignment allows researchers to understand the genomic structure and variation among genomes. Approaches based on direct pairwise comparisons of DNA sequences require large computational capacities. As a consequence, pipelines combining tools for orthologous gene identification and synteny have been developed.

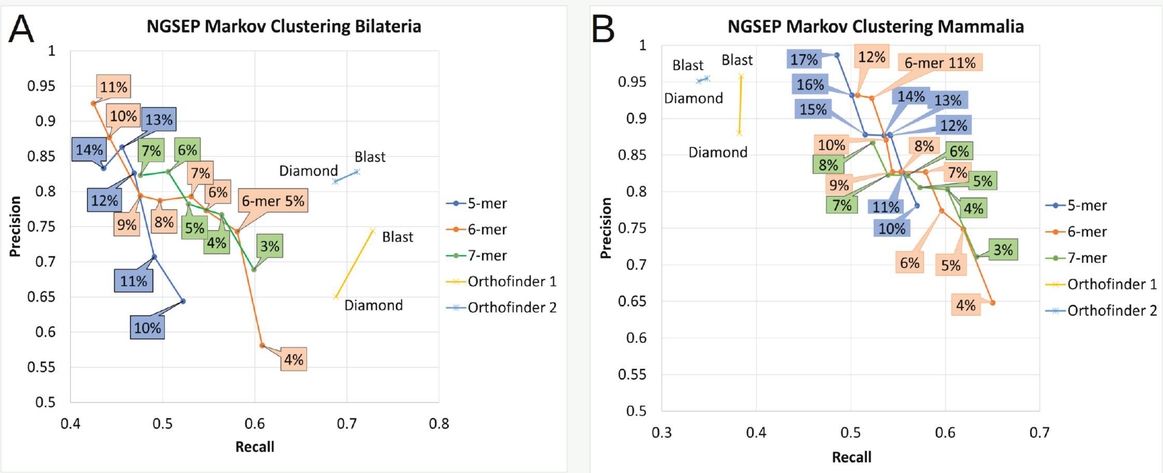
In this manuscript, we present the latest functionalities implemented in **NGSEP 4**, to identify orthogroups and perform whole genome alignments. NGSEP implements functionalities for identification of clusters of homologus genes, synteny analysis and whole genome alignment.

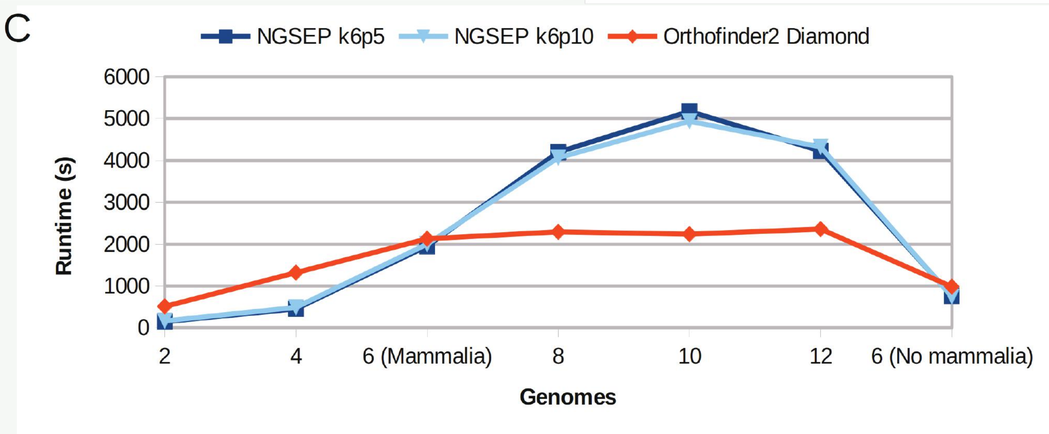
ALGORITHM



ORTHOGROUPS CLUSTERING

NGSEP clustering precision is stable to changes in taxonomy of Bilateria, while recall is better with narrow taxonomies.



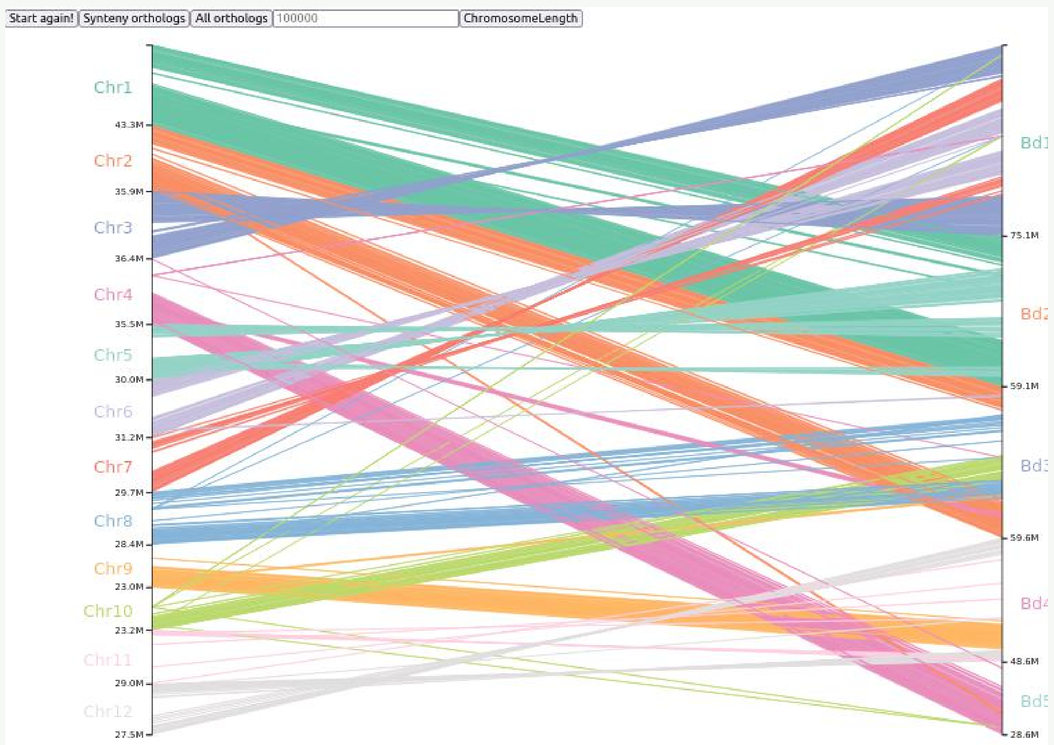


NGSEP is faster in the majority of the cases in comparison to Orthofinder2.

NGSEP clustering is a fast and accurate alignment-free method for orthologuos clustering. It is useful in a wide range of organisms.

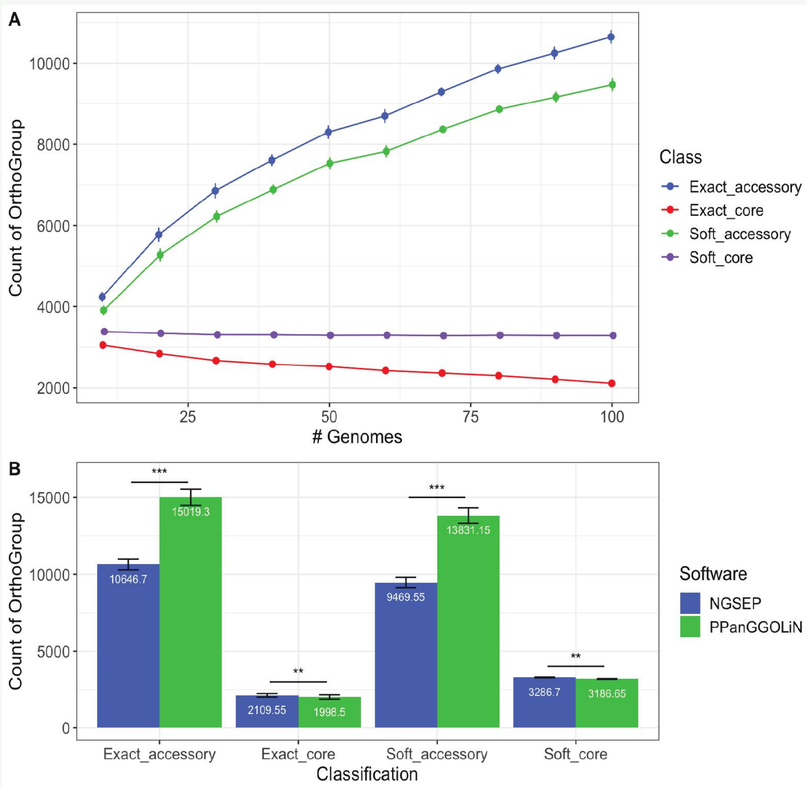
GENOME ALIGNMENT

NGSEP finds the collinear blocks between *Oryza sativa* and *Brachipodium distachion* chromosomes based on orthologous genes and their synteny



HTML visualization allows the interaction with blocks of genes within the chromosomes/contigs

PANGENOME ANALYSIS



NGSEP pangenome reconstruction over *Escherichia coli* public genomes shows a stable soft core-genome while the accessory-genome grows with the number of genomes.

NGSEP is faster and computing efficient than commonly used tools showing similar classification of gene families.

DISCUSSION & CONCLUSIONS

- The NGSEP algorithm for orthogroups identification has competitive accuracy and efficiency in comparison to commonly used tools. Even for large genomes, the NGSEP genomes aligner runs within minutes using one single core of a personal computer
- The implementation also includes a visualization of the whole genome alignment based on synteny of the orthogroups that were identified, and a reconstruction of the pangenome based on frequencies of the orthogroups among the genomes.
- NGSEP 4 also includes a new graphical user interface based on the JavaFX technology.
- We expect that these new developments will be very useful for several studies in evolutionary biology and population genomics.

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